

DATA SCIENCE (DS) — Complete Roadmap & Interview Index

B.Tech CSE 4th Year • Basic → Intermediate → Advanced • Interview + Online Assessment Preparation

How to use this index: Follow the numbered sequence. Each topic is intentionally listed as a future chapter/lesson rather than explained here. The roadmap covers learning order, practical skills, interview depth, coding/OA relevance, projects, deployment, and revision.

A. Common Prerequisites & Interview Core

Purpose: Shared foundation before specializing.

A.1 Programming Fundamentals — Variables, data types, operators, input/output, conditions, loops, functions, scope and recursion.

A.2 Python for Technical Interviews — Lists, tuples, sets, dictionaries, strings, slicing, comprehensions, functions, modules and exceptions.

A.3 OOP Essentials — Classes, objects, inheritance, polymorphism, abstraction, encapsulation and composition.

A.4 DSA Foundation — Arrays, strings, linked lists, stacks, queues, hash tables, trees, heaps, graphs and recursion.

A.5 Algorithms — Sorting, searching, two pointers, sliding window, prefix sums, greedy, divide-and-conquer and dynamic programming.

A.6 Complexity — Big-O, time/space complexity, amortized thinking and trade-offs.

A.7 SQL Basics — SELECT, filtering, joins, grouping, subqueries, CTEs, window functions and query optimization basics.

A.8 Git & GitHub — Repositories, branching, commits, merge/rebase, pull requests and collaborative workflow.

A.9 Linux & CLI — Files, permissions, processes, pipes, grep, sed, awk, package management and environment variables.

A.10 Interview Communication — Explain approach, assumptions, complexity, trade-offs, debugging and project decisions.

1. Data Science Foundations

Purpose: Understand the complete data-to-decision lifecycle.

1.1 What is Data Science? — Combines statistics, programming, domain knowledge and communication to solve data problems.

1.2 Data Science Lifecycle — Problem → data → cleaning → EDA → modeling → evaluation → communication → deployment.

1.3 Business Problem Framing — Objectives, stakeholders, KPIs, constraints and measurable success criteria.

1.4 Data Types — Numerical, categorical, ordinal, temporal, text, image and semi-structured data.

1.5 Data Quality — Accuracy, completeness, consistency, timeliness, uniqueness and validity.

2. Python for Data Science

Purpose: Master the language and ecosystem used in data work.

2.1 Python Core — Syntax, functions, collections, comprehensions, exceptions and modules.

2.2 NumPy — Arrays, shapes, indexing, broadcasting, vectorization and linear algebra operations.

2.3 pandas — Series, DataFrame, selection, filtering, groupby, merge, pivot and reshaping.

2.4 Data Cleaning — Missing values, duplicates, types, string cleaning and transformations.

2.5 Visualization — Matplotlib/Seaborn concepts, plots, annotations and chart selection.

2.6 Jupyter & Environments — Notebooks, virtual environments, packages and reproducible analysis.

3. SQL & Databases

Purpose: Essential for data retrieval and most data-science OAs.

3.1 SQL Fundamentals — SELECT, WHERE, ORDER BY, LIMIT, DISTINCT and expressions.

3.2 Aggregation — GROUP BY, HAVING and aggregate functions.

3.3 Joins — INNER, LEFT, RIGHT, FULL, CROSS and self joins.

3.4 Subqueries & CTEs — Nested queries, WITH clauses and readable transformations.

3.5 Window Functions — ROW_NUMBER, RANK, DENSE_RANK, LAG, LEAD and running aggregates.

3.6 Advanced SQL — CASE, date functions, string functions, set operations and NULL behavior.

3.7 Query Performance — Indexes, execution plans and avoiding expensive patterns.

4. Statistics

Purpose: Build rigorous reasoning from data.

4.1 Descriptive Statistics — Mean, median, mode, variance, standard deviation and percentiles.

4.2 Distributions — Normal, binomial, Poisson, uniform and skewed distributions.

4.3 Probability — Conditional probability, Bayes theorem and independence.

4.4 Sampling — Population, sample, sampling bias and central limit intuition.

4.5 Confidence Intervals — Point estimates, uncertainty and interpretation.

4.6 Hypothesis Testing — Null/alternative hypotheses, p-values, significance and errors.

4.7 Correlation — Pearson/Spearman, covariance and interpretation limitations.

4.8 A/B Testing — Experiment design, metrics, statistical significance and practical significance.

5. Exploratory Data Analysis

Purpose: Turn raw data into insights and hypotheses.

5.1 Univariate Analysis — Distributions, frequency, spread and outliers.

5.2 Bivariate Analysis — Relationships between pairs of variables.

5.3 Multivariate Analysis — Interactions, correlations and higher-dimensional patterns.

5.4 Outlier Analysis — IQR, z-score, robust methods and domain validation.

5.5 Missing Data Analysis — Patterns of missingness and suitable treatment.

5.6 EDA Storytelling — Convert observations into actionable questions and hypotheses.

6. Machine Learning for Data Science

Purpose: Use predictive models appropriately.

6.1 Regression — Linear, regularized and tree-based regression.

6.2 Classification — Logistic regression, trees, ensembles and SVM concepts.

6.3 Clustering — K-means, hierarchical clustering and DBSCAN.

6.4 Dimensionality Reduction — PCA and practical use cases.

6.5 Preprocessing — Encoding, scaling, feature engineering and pipelines.

6.6 Evaluation — Metrics, cross-validation, leakage and model selection.

6.7 Interpretability — Feature importance and explainability concepts.

7. Advanced Data Science

Purpose: Broaden from modeling into modern data products.

7.1 Time Series — Trend, seasonality, stationarity, lag features and forecasting.

7.2 NLP Basics — Tokenization, text cleaning, TF-IDF, embeddings and text classification.

7.3 Recommender Systems — Content-based, collaborative filtering and ranking.

7.4 Experimentation — A/B testing, causal thinking and experiment pitfalls.

7.5 Data Engineering Basics — ETL/ELT, batch/streaming, data warehouses, lakes and pipelines.

7.6 Big Data Concepts — Distributed processing, Spark concepts, partitions and joins.

7.7 Cloud Data Basics — Object storage, compute, managed databases and analytics services.

8. Data Visualization & Communication

Purpose: A strong data scientist must explain findings clearly.

8.1 Chart Selection — Match chart type to comparison, distribution, relationship or trend.

8.2 Dashboard Thinking — KPIs, filters, drill-downs, layout and stakeholder needs.

8.3 Storytelling — Context → insight → evidence → recommendation → impact.

8.4 Common Visualization Errors — Misleading scales, clutter, inappropriate charts and correlation overclaiming.

8.5 Presentation Skills — Explain technical findings to both technical and non-technical audiences.

9. Data Science Projects

Purpose: Create portfolio evidence across the full lifecycle.

9.1 EDA Project — Messy dataset → cleaning → analysis → visualization → insights.

9.2 Predictive Project — Problem framing → ML pipeline → evaluation → business interpretation.

9.3 Dashboard Project — SQL/data source → transformation → dashboard → stakeholder story.

9.4 End-to-End Project — Data ingestion → warehouse/processing → model → API/dashboard → documentation.

9.5 Project Defense — Why the data, method, metric, assumptions, limitations and next steps.

10. DS Interview & OA Mastery

Purpose: Prepare for common graduate hiring rounds.

10.1 Python OA — Collections, functions, strings, algorithms, debugging and complexity.

10.2 SQL OA — Joins, grouping, windows, CTEs, ranking and query optimization basics.

10.3 Statistics OA — Probability, distributions, estimation, testing and A/B testing.

10.4 ML Questions — Model choice, metrics, overfitting, preprocessing and evaluation.

10.5 Case Studies — Business problem → data → analysis → model → recommendation.

10.6 Product/Business Questions — KPIs, funnels, cohorts, retention, conversion and experiment design.

10.7 Behavioral Round — Project ownership, conflict, failure, learning and communication.

10.8 Final Mock Cycle — Timed OA → technical interview → case → project defense → HR.

FINAL INTERVIEW / OA FLOW

Phase 1: Fundamentals → **Phase 2:** Core concepts → **Phase 3:** Coding & problem solving → **Phase 4:** Tools/frameworks → **Phase 5:** Projects → **Phase 6:** System/design thinking → **Phase 7:** Mock OAs → **Phase 8:** Technical + HR interviews → **Phase 9:** Revision and company-specific preparation.

Recommended answer depth: Definition → intuition → syntax/formula → small example → edge cases → complexity/trade-offs → real-world use → common interview questions.

End goal: Be able to explain concepts without notes, implement core algorithms/models, debug code, interpret results, discuss project decisions, and solve timed OA questions.